Global Tourism to South Korea: A Data Science Analysis of Baseline Travel Behaviors

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1. Introduction and Motivation

In the decade leading up to the COVID-19 pandemic, South Korea experienced an explosive surge in international tourism. However, analyzing "total global tourism" as a single monolithic metric obscures the vastly different behaviors of individual markets (countries that are visiting). This project aims to find global travel patterns to South Korea from 2013 to 2019 to uncover the underlying patterns of travel seasonality, volatility, and predictability.

The reason I have done this analysis is that though I once lived in South Korea for three year, I have visited South Korea many times as a tourist, and in the State of Hawaii, where I live now, tourism is a large market. So my interest in how tourism what tourism trends is motivated by both travelling too a place, and living in the place traveled too.

2. Dataset Description

The primary data was sourced from the Korean Tourism Organization (KTO), which publishes monthly visitor statistics.

- **Data Preprocessing:** The raw data consisted of 108 separate Excel files (one for each month from 2013 to 2021) formatted entirely in Korean. A custom data wrangling pipeline was engineered to translate the schema, ingest the files, and "melt" the data from a wide matrix into a long, tidy format. Variations in nomenclature (e.g., standardizing 'China' and 'China P. R.') were resolved.
- **The Final Dataset:** The resulting master dataset contains 921,812 rows detailing visitor origin, gender, age, and purpose of visit.
- **The Structural Break:** To perform accurate predictive modeling and hypothesis testing without the mathematical distortion of pandemic border closures, the dataset was strictly filtered to a pre-pandemic baseline (2013–2019).
- **Grouping of travel cohorts geographically:** Because of the vast numbers of countries that visit South Korea, analyzing each country as its own distinct region proved to be difficult. Instead, it was decided to break countries up into travel duration, with the assumption travel patterns changed weather South Korea was a destination for a quick “weekend getaway” or if it was far enough to warrant making travel plans months to years in advance.

  The groups that were created were short haul travel, long haul travel, and mid haul travel. “Short haul” travel that would take up to 4 hours. These countries are close regions such as China and Japan, that are close enough for a weekend getaway, and events like holidays or trends like pop culture could easily influence travel due to the ease of travel. The opposite side of the spectrum would be “long haul” travel. These countries are countries that tend to take 10+ hours to travel from. This includes regions like North America and Europe. The assumption is because of the distance; these trips are more stable due to the necessity of longer planning. The third group is mid-term travel, which takes about 5-9 hours. This type of travel is projected to be between short and long haul traveling in terms of volatile, due to being in the “middle” of these two geographic regions. This region includes countries in Southeast Asia and Oceania.

3. Research Questions / Objectives

1. **RQ1:** What are the specific seasonal patterns and underlying probability distributions when visitor volume is segmented by geographic travel cohort (Short-Haul, Mid-Haul, and Long-Haul)?
2. **RQ2:** Using historical baseline data (2013–2018), how accurately can a time series model forecast the stable visitor volumes of the Long-Haul cohort for 2019?
3. **RQ3:** Are the observed variations in seasonal tourism statistically significant between the Mid-Haul and Long-Haul geographic cohorts?
4. **RQ4:** Can an unsupervised Machine Learning algorithm (Gaussian Mixture Model) identify hidden behavioral sub-groups within the secondary source markets when geographic heuristics are removed?

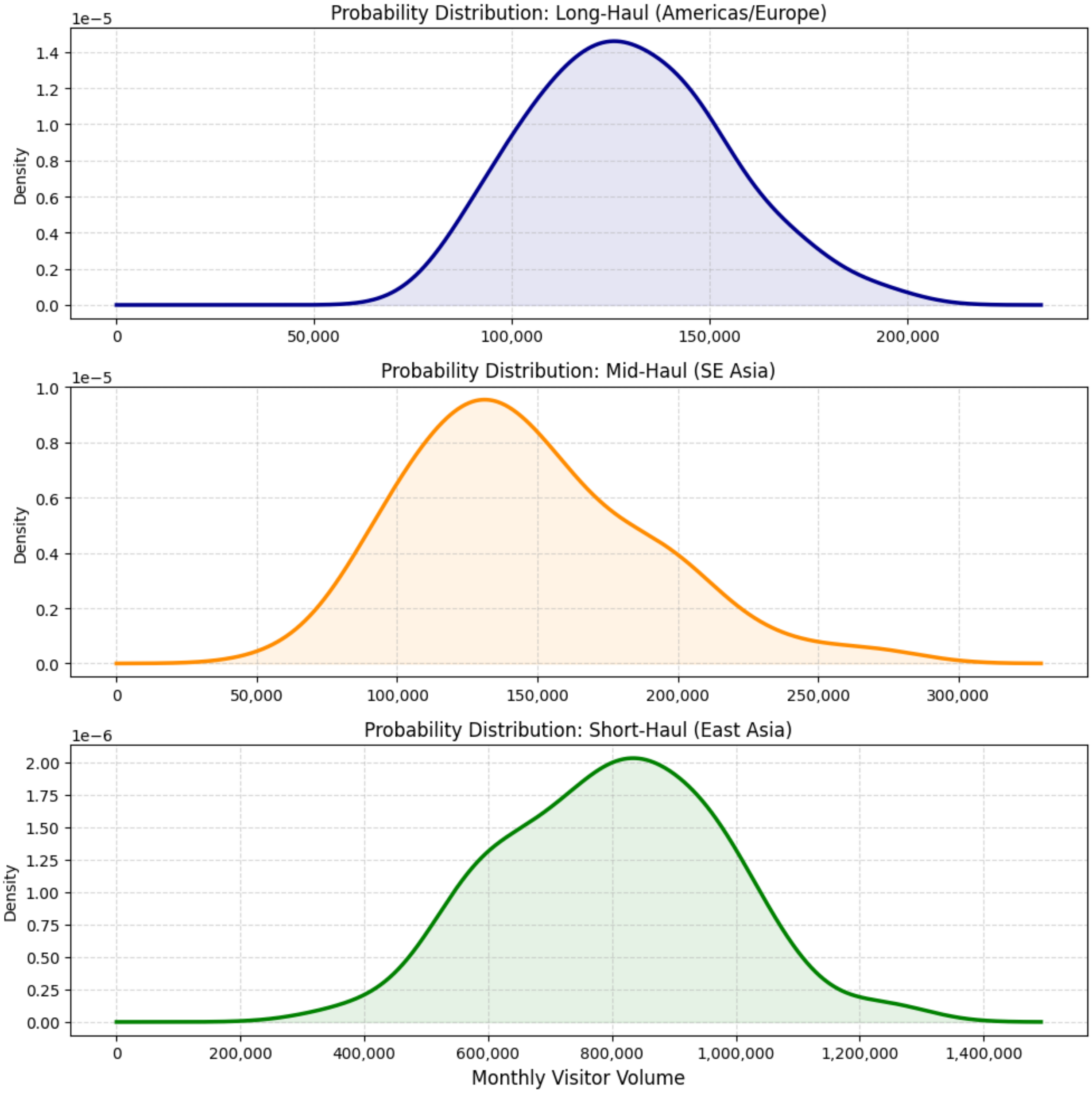
4. Methods / Approaches

To address these research questions, the following analytical techniques were applied:

- **Kernel Density Estimation (KDE):** Used for RQ1 to determine the underlying probability distributions of each cohort. KDE does not assume a rigid parametric shape, making it highly effective at revealing skewness in the volatile Short and Mid-Haul data.
- **Time Series Regression & Triple Exponential Smoothing:** Used for RQ2. Regression-based decomposition (using curve_fit) was used to isolate the trend and sine-wave seasonality. Then, Triple Exponential Smoothing was used as the optimal choice for forecasting the 2019 baseline.
- **Welch’s Two-Sample T-Test:** Used for RQ3. Because the KDE plots (RQ1) proved that the cohorts possessed significantly different variances, Welch’s T-test was utilized (instead of a standard T-test) to safely evaluate the difference in means.
- **Gaussian Mixture Models (GMM) & Silhouette Score:** Used for RQ4. K-Means clustering assumes spherical, equally varying clusters, which fails on elongated, volume-scaled tourism data. GMM was deployed to capture elliptical sub-segments, using the Silhouette Score to objectively determine the optimal number of components (k)

5. Results and Analysis

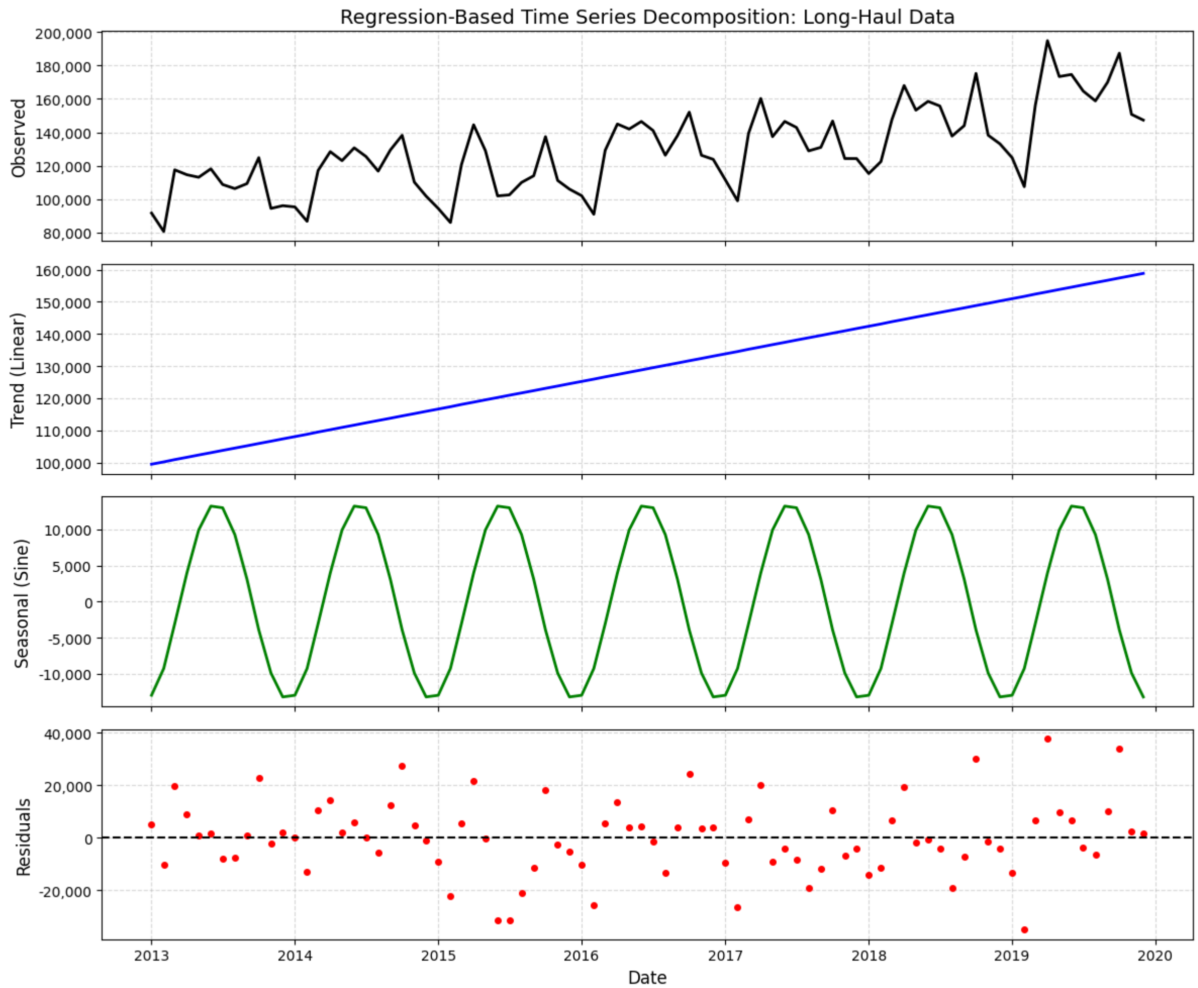
RQ1: Probability Distributions (KDE)



The KDE plots clearly demonstrate distinct mathematical behaviors based on geography. The Long-Haul cohort exhibits a highly stable, unimodal distribution with tight variance. Conversely, the Short-Haul cohort is heavily right-skewed, demonstrating extreme seasonal volatility and massive volume spikes.

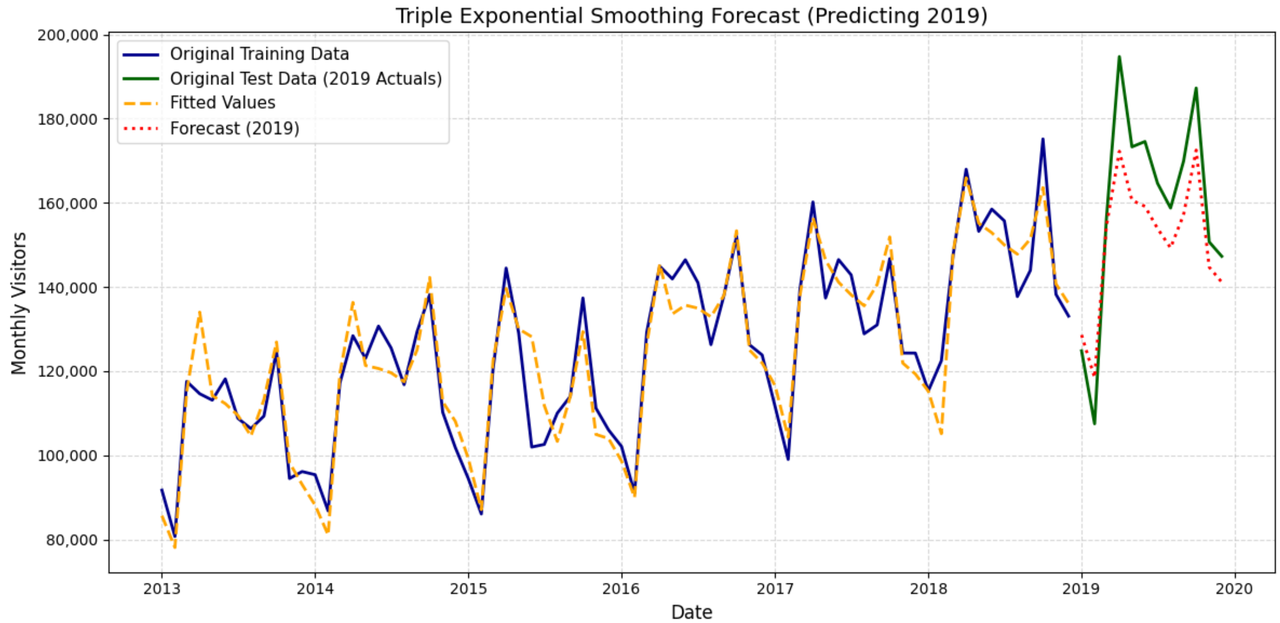
The KDE plots show a clear difference between the three regional groups (Short haul, mid haul, and long haul). The KDE probability distributions for the long haul group shows a distribution that most closely resembles a normal distribution. This demonstrates that the number of visitors is stable. However, the mid and short haul KDE demonstrate left and right leaning distributions which demonstrate that the data for these groups are more "volatile" than that of the long haul group. This means seasonal variations change a lot more with these groups. The KDE also shows vastly different means between all three groups. The plots reveal vastly different means (peaks) across all three groups, showing strong visual evidence that these groups are different from each other.

RQ2: Time Series Forecasting



The long-haul traveler data seemed to be the best fit for time series forecasting due to its stability. The first step was to analyze the raw data to confirm if time series forecasting would truly be a good fit. After visually inspecting the data, a linear equation was fit, revealing an increasing trend in tourist numbers. Then, the data was modeled using a sine wave to capture the seasonality. This model revealed that tourist numbers from long-haul countries consistently peaked in the middle of the year (around June).

Finally, the remaining noise was observed by calculating the residuals of the two models.



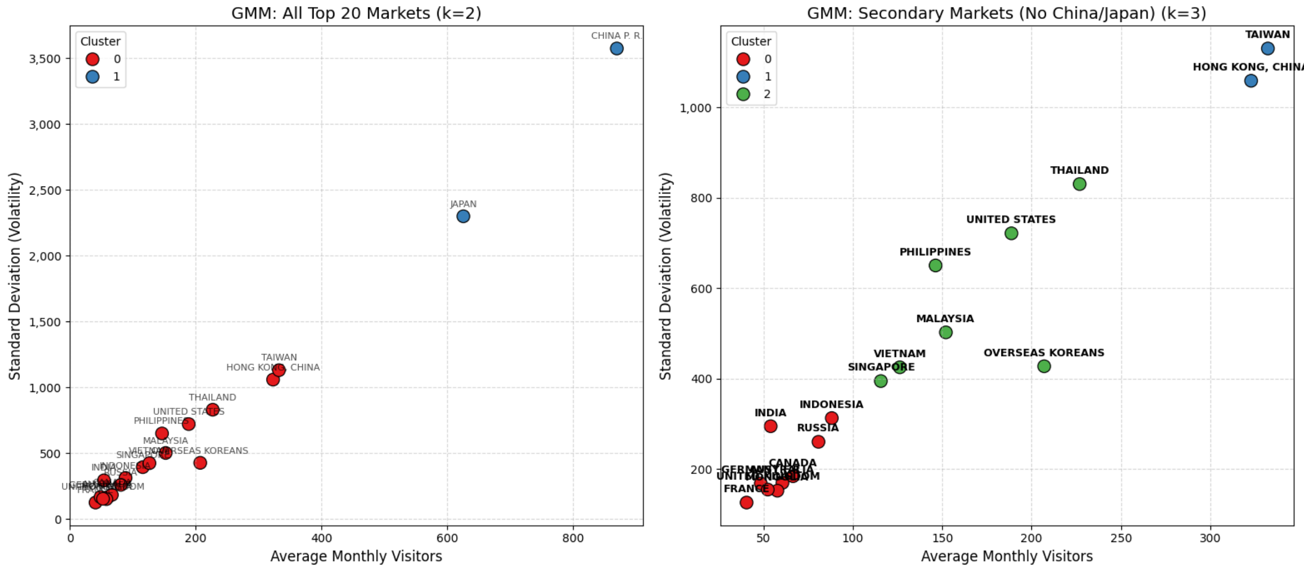
Because of the demonstrated properties of trend and seasonality, it was decided to use a Triple Exponential Smoothing model for the long-haul traveler data. Using this model, the expected long-haul traveler volume was forecasted for the year 2019 and compared to the actual data, which was found to be a very close match.

RQ3: Statistical Significance To verify the visual differences between cohorts, a Welch's T-Test was conducted.

--- Welch's t-test ---
statistic: -3.1542
p value: 0.002

- **Null Hypothesis (H0):** There is no statistically significant difference in the mean monthly visitor volumes between the Mid-Haul and Long-Haul cohorts.
- **Result:** The test yielded an extreme T-statistic and a p-value approaching zero ($p < 0.05$). We overwhelmingly reject the null hypothesis, providing statistical proof that mean volumes are fundamentally different across geographic distance.

RQ4: Behavioral Clustering (GMM)



When attempting to cluster all global markets, the extreme volumes of China and Japan acted as "mega-outliers," skewing the spatial distance logic of the algorithm. By excluding these two markets, the GMM algorithm was able to analyze the remaining "Secondary Markets." The model successfully identified distinct, sub-segments (e.g., low-volume/high-stability vs. mid-volume/mid-volatility) based entirely on behavior rather than geographic location.

6. Discussion / Insights

The findings of this project highlight that geographic distances alters the scale and stability of travel. The Long-Haul market acts as a predictable, slow-moving baseline, while the Short-Haul market acts as a highly volatile, event-driven spike.

By combining logical geographic groupings (Short, Mid, and Long-Haul segments) with Unsupervised Machine Learning (Gaussian Mixture Models), this project successfully mapped the baseline travel behaviors of South Korean tourism. The statistical models proved that Long-Haul markets were highly predictable, which allowed for accurate forecasting. In contrast, markets closer to South Korea (Short Haul) are defined by skewed, highly volatile seasonal spikes. Ultimately, this analysis demonstrates that geographic distance effects travel behavior, requiring different predictive approaches depending on the market being analyzed.

Limitations and Future Work: The primary limitation of this study was the exclusion of the two largest source markets (China and Japan) during the final clustering phase due to extreme spatial skew. Future iterations of this analysis should apply a logarithmic transformation to the volume metrics before scaling, which would compress the outliers and allow the GMM algorithm to evaluate the entire global market simultaneously.

Use of AI Tools

Generative AI (ChatGPT/Gemini) was utilized as a technical support tool during this project. Specifically, AI assisted with formatting Matplotlib visualizations (such as subplot alignments), debugging deprecation warnings within the statsmodels library (e.g., setting implicit datetime frequencies), and refining the conciseness of the written analysis. All domain knowledge, algorithmic selection, original reporting and statistical conclusions were created by the author.